

Muhammad Maawaz

Kahuta, Rawalpindi, Pakistan

✉ maawazali111@gmail.com



+92-346-5120975



github.com/mwchuchu



linkedin.com/in/maawaz



maawaz.vercel.app

AI | Machine Learning | Deep Learning | NLP | LLMs | RAG | Transformers | Embeddings | Vector Databases | FAISS | Python | PyTorch | TensorFlow | Scikit-learn | FastAPI | Flask | REST APIs | MLOps | Model Deployment | Backend Engineering | Data Science | Computer Vision | Transfer Learning | Predictive Modeling | Model Evaluation | React Native | JavaScript | C++ | SQL | Pandas | NumPy | Git | GitHub | Docker

Professional Summary

Computer Science graduates specialized in Machine Learning, Artificial Intelligence, and Natural Language Processing. Experienced in building end-to-end AI systems including RAG pipelines, vector database search systems, and intelligent chatbot applications. Strong focus on scalable backend systems, applied ML models, and production-ready AI solutions.

Technical Skills

Programming Languages: Python, JavaScript, C++

Machine Learning: Scikit-learn, PyTorch, TensorFlow, Deep Learning, Model Evaluation

Natural Language Processing: Transformers, LLMs, Embeddings, RAG Pipelines, Sentence Transformers

Backend Development: FastAPI, Flask, REST APIs, Celery

Databases: MongoDB, SQL, FAISS (Vector Database)

Tools & Platforms: Git, GitHub, Postman, VS Code, Docker (basic)

Frontend: React Native, HTML, CSS, JavaScript

Deployment & MLOps: FastAPI and Flask, REST API integration, Docker, Production-ready ML pipeline development

Projects

AI-Based Skill Game (Mobile AI Application)

- Developed AI-powered quiz generation app using Gemini Flash 1.5 API
- Implemented adaptive difficulty levels (Beginner, Intermediate, Advanced)
- Built React Native frontend with MongoDB backend integration

Enhanced Smart Legal Clause Analyzer (RAG System)

- Designed Retrieval-Augmented Generation pipeline using FAISS vector database
- Implemented document chunking, embedding generation, and semantic search
- Built scalable backend using FastAPI and asynchronous processing with Celery

AI Automation Customer Support Agent

- Built LLM-based customer support system for multiple query resolutions with context-aware responses and semantic knowledge retrieval.
- Developed FastAPI-based ticket classification and routing engine, reducing manual support workload by 60–80%.
- Integrated Gemini APIs for real-time, scalable conversational AI automation.

Video-to-Quiz AI Learning Platform

- Built system to convert lecture videos into MCQs and True/False questions
- Used Deepgram API for speech-to-text transcription
- Integrated Gemini API for automated question generation and evaluation
- Developed chatbot for contextual question answering from video content

Spam Message Detection System

- Built ML classification system using Dense, LSTM, and Bi-LSTM models
- Evaluated model performance for accuracy and generalization
- Deployed model using Flask with web interface integration

PDF Chatbot (RAG-Based System)

- Built multi-document chatbot using embedding-based retrieval system
- Implemented semantic search over large PDF datasets
- Optimized pipeline for fast and relevant response generation

Experience

Machine Learning Intern — Dive Deep AI (Islamabad)

July 2025 – August 2025

- Built a Retrieval-Augmented Generation (RAG) based legal clause chatbot using FAISS for accurate semantic query retrieval
- Implemented transfer learning models including MobileNetV2 and ResNet101 for Pokémon image classification
- Achieved 90%+ classification accuracy through model optimization and fine-tuning

Data Science Intern — Developer's Hub (Remote)

September 2024 – December 2024

- Designed and deployed machine learning models integrated into production-level web applications
- Improved system decision-making efficiency by 30% through optimized predictive modeling
- Worked on end-to-end ML pipelines from data preprocessing to deployment

Education

Bachelor of Science in Computer Science (BSCS)

Bahria University Islamabad

(October 2021- July 2025)

Certifications

- Machine Learning with Python — freeCodeCamp
- Full Stack AI Engineer - Udemy
- Mastering Pytorch - Udemy
- Certified AI engineering Master class - Udemy
- Tensor flow master class - Udemy
- Azure Devops - Udemy

Areas of Interest

Machine Learning Engineering, Natural Language Processing, Retrieval-Augmented Generation (RAG), AI Automation Systems, Intelligent EdTech Platforms